

Heredity — MCQ Worksheet

Science • Chapter 8 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The father of genetics is:

- (A) Charles Darwin
- (B) Gregor Mendel
- (C) Watson and Crick
- (D) Robert Hooke

Q2. Mendel performed his experiments on:

- (A) Mice
- (B) Pea plants
- (C) Drosophila
- (D) Bacteria

Q3. A unit of inheritance is called a:

- (A) Chromosome
- (B) Gene
- (C) Protein
- (D) Cell

Q4. Alternative forms of a gene are called:

- (A) Alleles
- (B) Loci
- (C) Phenotypes
- (D) Mutations

Q5. In a monohybrid cross of $TT \times tt$, the F_1 generation is:

- (A) All TT
- (B) All tt
- (C) All Tt
- (D) TT , Tt and tt in equal numbers

Q6. In a monohybrid cross, the F_2 phenotypic ratio is:

- (A) 1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 2:1

Q7. In a monohybrid cross the F_2 genotypic ratio is:

- (A) 1:2:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:1

Q8. In a dihybrid cross the F_2 phenotypic ratio is:

- (A) 3:1
- (B) 1:2:1
- (C) 9:3:3:1
- (D) 1:1:1:1

Q9. The trait that appears in F_1 generation is called:

- (A) Recessive
- (B) Dominant
- (C) Co-dominant
- (D) Incomplete

Q10. Mendel's Law of Segregation states that:

- (A) Alleles separate during gamete formation
- (B) Genes never separate
- (C) Only dominant alleles pass
- (D) Sex determines all traits

Q11. Number of chromosomes in a normal human cell is:

- (A) 22
- (B) 23 pairs (46)
- (C) 24 pairs (48)
- (D) 20 pairs (40)

Q12. The sex chromosomes in a human female are:

- (A) XY
- (B) XX
- (C) XO
- (D) YY

Q13. The sex of a human child is determined by:

- (A) The mother
- (B) The father
- (C) Both equally
- (D) The environment

Q14. A human male produces sperms with:

- (A) Only X chromosome
- (B) Only Y chromosome
- (C) Either X or Y chromosome
- (D) Both X and Y in same sperm

Q15. Probability of having a male child in humans is:

- (A) 1/4
- (C) 1/2

- (B) 1/3
- (D) 2/3

Q16. Acquired traits are NOT passed on to offspring because:

- (A) They are too weak
- (B) They do not affect DNA of germ cells
- (C) Only humans pass traits
- (D) They are recessive

Q17. Variations are useful because:

- (A) They keep populations identical
- (B) They allow species to survive in changing environments
- (C) They are always harmful
- (D) They cause mutations only

Q18. In Mendel's pea cross, the recessive trait reappears in:

- (A) F1 generation
- (B) F2 generation
- (C) Both F1 and F2
- (D) Never

Q19. If pure-breeding round (RR) yellow seeds are crossed with wrinkled (rr) green ones, F1 will be:

- (A) All wrinkled green
- (B) All round yellow
- (C) 50% round, 50% wrinkled
- (D) All round green

Q20. Genes are located on:

- (A) Chromosomes
- (B) Cell membrane
- (C) Cytoplasm only
- (D) Mitochondria only

Q21. Number of autosomes in a human cell is:

- (A) 22 pairs
- (B) 23 pairs
- (C) 21 pairs
- (D) 20 pairs

Q22. Variation in an organism arises because of:

- (A) Imperfect DNA copying and sexual recombination
- (B) Only diet
- (C) Only environment
- (D) No genetic factors

Q23. In sexual reproduction, the chromosomes from each parent are present in the gamete in:

- (A) Half the number
- (B) The full number
- (C) Double the number
- (D) Random number

Q24. A homozygous tall pea plant has alleles:

- (A) Tt
- (B) TT
- (C) tt
- (D) Tt or TT

Q25. In Mendel's experiments, the parental generation is called:

- (A) F0
- (B) F1
- (C) F2
- (D) P generation